

JavaScript Basics – Mini Project Assignment

You are asked to create a simple **Student Management System**. This project will help you practice variables, data types, operators, conditions, loops, functions, objects, arrays and array methods.

Initial Data

```
students = [ {name:'Ahmed', age:20, grade:14}, {name:'Sara', age:22, grade:18}, {name:'Ali', age:19, grade:9}, {name:'Yasmine', age:21, grade:16} ]
```

Task 1

Display all students using a loop.

Task 2

Determine whether each student passed (grade >= 10) using if...else.

Task 3

Count how many students passed.

Task 4

Find the student with the highest grade.

Task 5

Create a function isAdult(age) that returns true if age >= 18 and false otherwise.

Task 6

Use find() to find Sara and filter() to get all students who passed.

Bonus

Add a new student, print the new array length, sort students by grade from highest to lowest.

Submission: Submit a single JavaScript file containing your solution and clear console outputs.